NTNU – Trondheim
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Science and Technology

TTT4120 Digital Signal Processing
Fall 2025

Lecture: Discrete-Time Signals in Time-Domain

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1

Lecture in course book*

- Proakis, Manolakis Digital Signal Processing, 4th Ed.
  - 1.1 Signals, systems and signal processing
  - 1.2 Classification of signals
  - 1.3 The concept of frequency in continuous-time and…
  - 1.4.1 Sampling of analog signals
  - 2.1 Discrete-time signals

*Level of detail is defined by lectures and problem sets

2

2

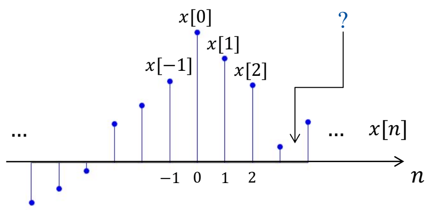
Contents and learning outcomes

- Discrete-time signals
- Power of digital signal processing (DSP)
- Properties, classification, and manipulations of sequences
- A few typical sequences
- Discrete-time sinusoids and sampling of continuous-time sinusoids

Discrete-time signals

- Discrete-time versus continuous-time?

Discrete-time signals…



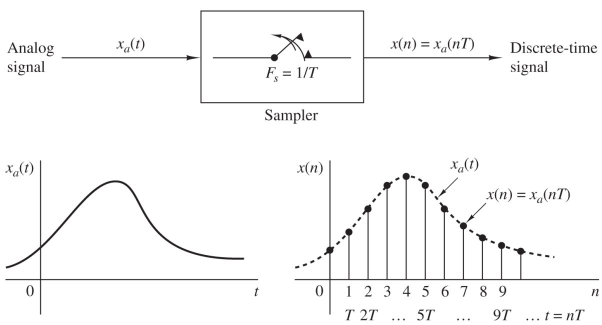
- A discrete-time signal $x[n]$ is represented by a sequence of numbers
- Sequence $x[n]$ can represent a discrete-time signal, where each number $x[n]$ corresponds to a signal amplitude at instant n

$$x[n] = \{\dots\ x[-2], x[-1], \underline{x[0]}, x[1], \dots\}$$

5

5

Discrete-time signals…



- Often $x[n]$ is obtained by uniform *sampling* of an analog signal $x_a(t)$

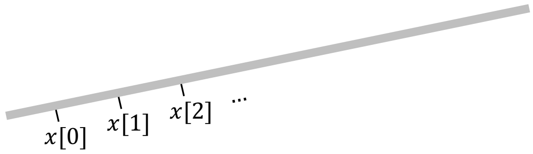
$$x[n] \triangleq x_a(nT)$$

- Sampling period $T = \frac{1}{F_s}$ (sampling rate F_s)

6

6

Discrete-time signals…

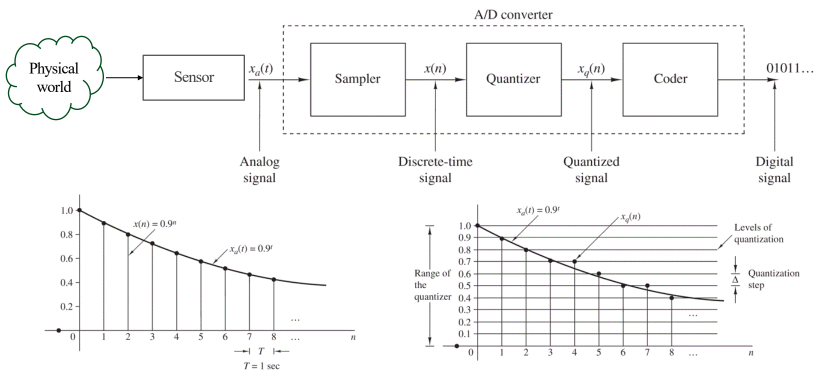


- Note that the ‘sampling interval’ T need not be time
- For example, if $x_a(t)$ is the temperature along a metal rod, then if T is a length unit, $x[n] \triangleq x_a(nT)$ represents the temperature at uniformly placed sensors along this rod
- Different choices of T lead to different sequences

7

7

Characterization of signals

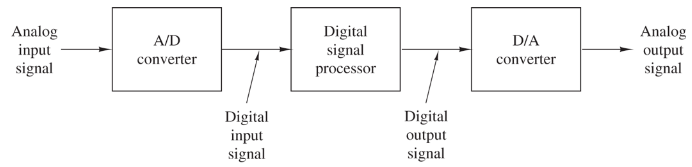


- Analog signal $x_a(t)$: continuous in time and amplitude
- Sampled-data signal $x[n]$: discrete-time and continuous-amplitude
- Digital signal $x_q[n]$: discrete in time and amplitude (quantized)

8

8

Power of DSP



- Digital signal
  - Discrete-time and discrete-valued sequence of numbers (last attribute less essential for DSP basics)
- Digital signal processing
  - Sequence is transformed to another sequence by means of arithmetic operations

9

9

Power of DSP…

- Analog signal processing:
  - processes continuous-time signals
  - often modeled by differential equations and analog filters
- Digital signal processing:
  - processes discrete-time sequences of numbers on digital hardware or in software
  - often modeled by difference equations and digital filters

Once a signal is in digital form, we can implement a wide range of algorithms, subject to *sampling rate*, *quantization/word length* (numerical precision), and *latency* constraints.

10

10

Operations on discrete-time signals

- Amplitude scaling, addition, and multiplication of sequences

$$y[n] = ax[n]$$

$$y[n] = x_1[n] + x_2[n]$$

$$y[n] = x_1[n]x_2[n]$$

- Time shifts and folding

$$y[n] = x[n-k]$$

$$y[n] = x[-n]$$

- Time shifts plus folding

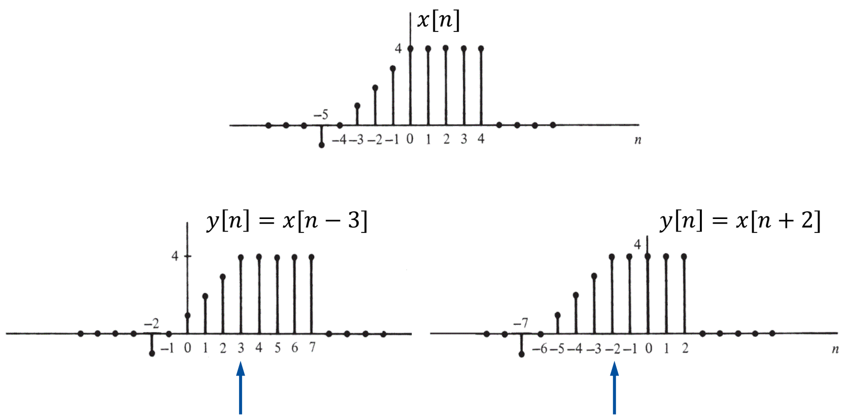
$$y[n] = x[-n+k]$$

11

11

Operations on discrete-time signals…

- Example (time-shift): Given $x[n]$ below, plot $x[n-3]$ and $x[n+2]$

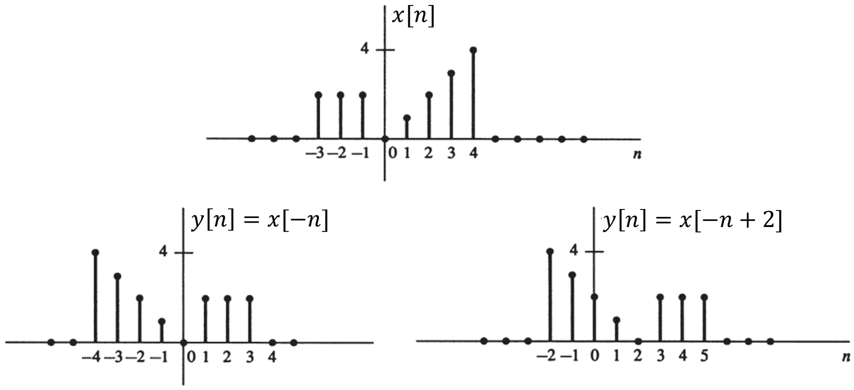


12

12

Operations on discrete-time signals…

- Example (folding): Given $x[n]$ below, plot $x[-n]$ and $x[-n+2]$



Basic properties of discrete-time signals

- A sequence $x[n]$ is causal if

$$x[n] = 0, n < 0$$

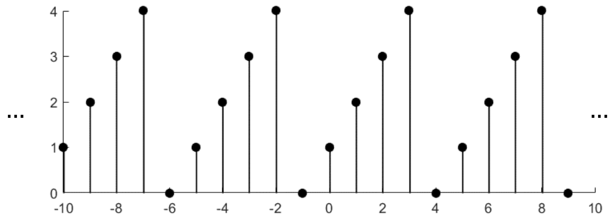
- A sequence $x[n]$ is periodic with period N for all n if

$$x[n+N] = x[n], \forall n$$

where smallest N satisfying the above is the fundamental period

- A sequence that is not periodic is called non-periodic or aperiodic

Basic properties of discrete-time signals…



- Is the above sequence periodic?
- If so, what is the fundamental period?

Basic properties of discrete-time signals…

- A real-valued sequence $x_{\mathrm{e}}[n]$ is called even if

$$x_{\mathrm{e}}[n] = x_{\mathrm{e}}[-n], \forall n$$

- A real-valued sequence $x_{\mathrm{o}}[n]$ is called odd if

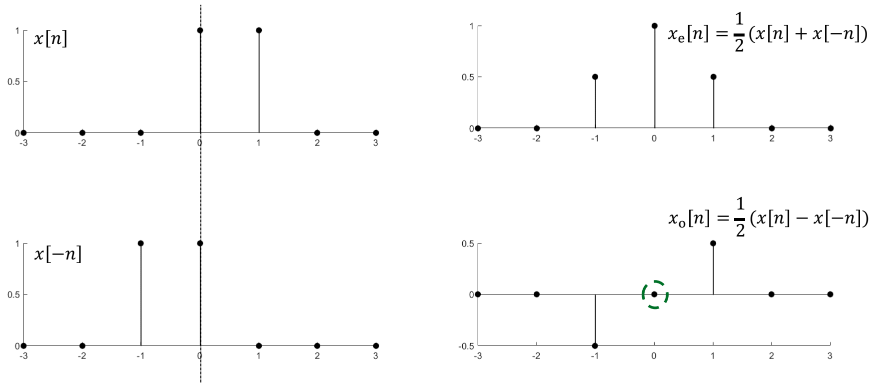
$$x[-n] = -x[n], \forall n$$

- Any real-valued sequence can be expressed as

$$x[n] = x_{\mathrm{e}}[n] + x_{\mathrm{o}}[n]$$

$$x_{\mathrm{e}}[n] = \frac{1}{2}(x[n] + x[-n]) \qquad x_{\mathrm{o}}[n] = \frac{1}{2}(x[n] - x[-n])$$

Basic properties of discrete-time signals…



17

17

Classifications of discrete-time signals

- A sequence is bounded if $|x[n]| \leq B_x < \infty$ for all n
- A sequence is absolutely summable if

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

- A sequence is square-summable if its energy

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty$$

is bounded. Such signal is called an energy signal

18

18

Classifications of discrete-time signals…

- Not all sequences are energy signals (e.g., periodic signals)
- Average power of sequence $x[n]$ is defined as

$$P_x = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$$

- If P_x is finite, the signal is called a power signal

19

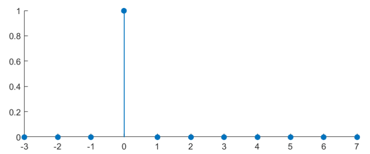
19

Basic sequences

- Unit impulse:

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$

$\delta[n]$

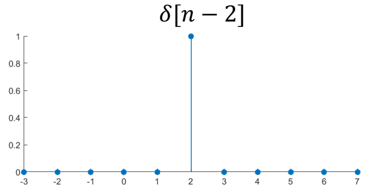


- Delayed unit impulse:

$$\delta[n-k] = \begin{cases} 1 & n = k \\ 0 & n \neq k \end{cases}$$

Matlab
```
k  = 2;
n = (-3:7)
delta = [(n-k)==0];
stem(n,delta)
```

$\delta[n-2]$



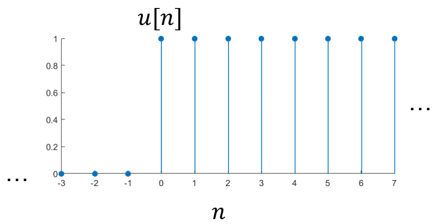
20

20

Basic sequences…

- Unit step:

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



Matlab
```
n = (-3:7)
u = [n>=0];
stem(n,u)
```

21

21

Basic sequences…

- Relationship between $u[n]$ and $\delta[n]$:
  - Unit impulse is the first-order difference of the unit step

$$\delta[n] = u[n] - u[n-1]$$

  - Unit step is the running sum of the unit impulse

$$u[n] = \sum_{k=-\infty}^{n} \delta[k]$$

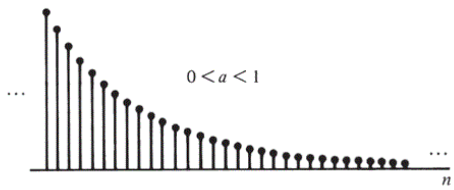
22

22

Basic sequences…

- Real-valued exponential function

$$x[n] = a^n, \forall n \text{ and } a \in \mathbb{R}$$

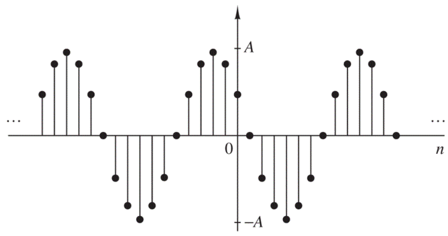


- What if a is complex-valued, i.e., $a \in \mathbb{C}$?

23

23

Discrete-time sinusoid



$$x[n] = A\cos[\omega n + \theta] = A\cos[2\pi f n + \theta]$$
$$= \frac{A}{2}\left(e^{j[\omega n+\theta]} + e^{-j[\omega n+\theta]}\right)$$

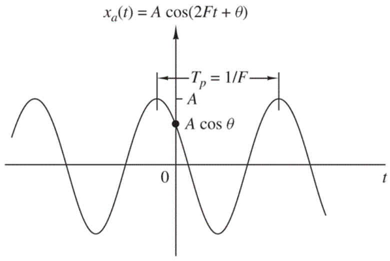
- What about the notion of frequency in discrete time?
- What about the notion of periodicity for discrete-time sinusoids?

24

24

Discrete-time sinusoid…

- Continuous-time sinusoid:



- Consider two signals

$$x_1(t) = A\cos(\Omega_1 t) = A\cos(2\pi F_1 t)$$

$$x_2(t) = A\cos(\Omega_2 t) = A\cos(2\pi F_2 t)$$

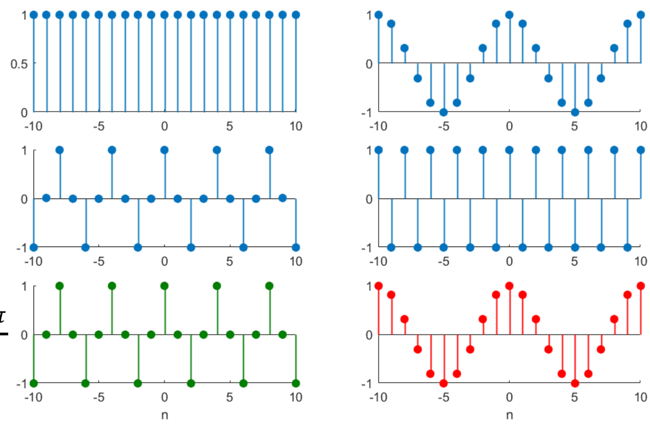
where $F_2 > F_1$
  - Signal $x_2(t)$ will oscillate faster than $x_1(t)$
  - In general $x_2(t) \neq x_1(t)$, except at some possible points

25

25

Discrete-time sinusoid…

- Digital frequency: $x[n] = \cos[\omega n] = \cos[2\pi f n]$



$\omega = 0$, $f = 0$

$\omega = \frac{\pi}{5}$, $f = \frac{1}{10}$

$\omega = \frac{\pi}{2}$, $f = \frac{1}{4}$

$\omega = \pi$, $f = \frac{1}{2}$

$\omega = \frac{3\pi}{2}$, $f = \frac{3}{4}$

$\omega = \frac{9\pi}{5}$, $f = \frac{9}{10}$

26

26

Discrete-time sinusoid…

- Discrete-time sinusoid is 2π-periodic in frequency

$$\cos[(\omega + 2k\pi)n] = \cos[\omega n + 2kn\pi] = \cos[\omega n]$$

$\Longrightarrow$ Any sinusoidal sequence with $|\omega| > \pi$ is identical to a sinusoidal sequence with $|\omega| \leq \pi$!

  – Verify this for the green and red sinusoids in previous slide

- Lowest frequency at $\omega_k = 0 + 2\pi k$
- Highest frequency at $\omega_k = \pi + 2\pi k$

$\Longrightarrow$ Range of frequencies is finite

$$-\pi \leq \omega \leq \pi, \text{ or } -\frac{1}{2} \leq f \leq \frac{1}{2}$$

$$(0 \leq \omega \leq 2\pi, \text{ or } 0 \leq f \leq 1)$$

27

27

Discrete-time sinusoid…

- Is a discrete-time sinusoid a periodic sequence?

$$x[n] = x[n + N]?$$

$$\cos[2\pi f\, n] = \cos[2\pi f(n + N)]?$$

- Answer: (Yes/No/Sometimes) [Tick your option]

28

28

Discrete-time sinusoid…

- Answer: Sometimes

$$\cos[2\pi n] = \cos[2\pi f(n + N)]$$
$$\Rightarrow 2\pi f N = 2\pi k$$
$$\boxed{\Rightarrow f = \frac{k}{N}}$$

- A discrete-time sinusoid is periodic *iff* its frequency f is rational in its lowest terms; fundamental period N

29

29

Discrete-time sinusoid…

- Example: Determine if the discrete-time signals below are periodic; if they are, determine their periods

1. $x[n] = \cos\left[\frac{12\pi}{5} n\right]$
2. $x[n] = \cos[0.02n + 3]$

30

30

Complex exponential

- Complex exponential: $x[n] = Ae^{j[2\pi fn+\theta]}$
- Same properties as discrete-time sinusoids
  - 2π-periodic in (angular) frequency
  - Periodic sequence if frequency f is rational
- Building block Fourier representations (DTFT/DFS/DFT)

31

31

Sampling a sinusoidal signal

- Consider sampling sinusoidal signal at intervals $nT = n/F_s$

$$x_a(t) = A\cos(\Omega t) = A\cos(2\pi Ft)$$

- Discretized signal

$$x[n] = x_a(nT) = A\cos\left[2\pi \frac{F}{F_s} n\right] = A\cos[2\pi fn]$$

$$\Rightarrow f = \frac{F}{F_s} \text{ or } \omega = \Omega T \text{ (relative/normalized frequency)}$$

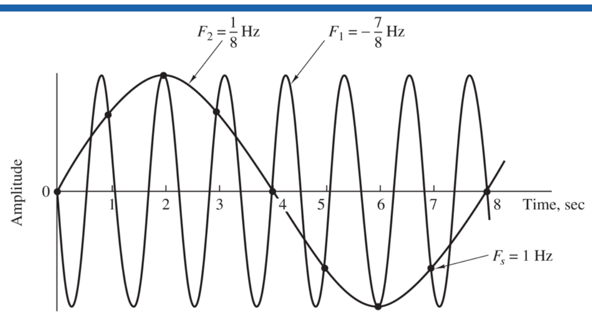
- For accurate representation we know from before

$$-\frac{1}{2} \le f \le \frac{1}{2} \quad \Leftrightarrow \quad -\frac{F_s}{2} \le F \le \frac{F_s}{2}$$

32

32

Aliasing



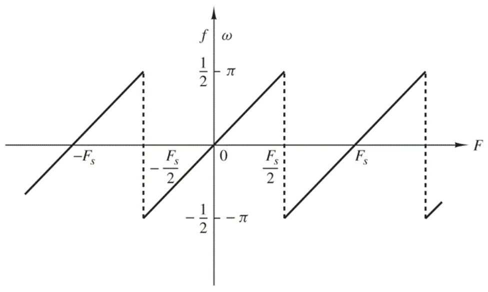
$$A\cos\left[2\pi\frac{F_2}{F_s}n\right] = A\cos\left[2\pi\frac{1}{8}n\right]$$

$$A\cos\left[2\pi\frac{F_1}{F_s}n\right] = A\cos\left[2\pi\frac{(-7)}{8}n\right] = A\cos\left[2\pi\underbrace{\left(\frac{(-7)}{8}+1\right)}_{=\frac{1}{8}}n\right]$$

33

33

Aliasing…



- Discrete-time versus continuous-time frequency variables in periodic sampling

34

34

Summary

- Today we discussed:
  - Discrete-time signals in time-domain
- Next:
  - Discrete-time systems in time-domain

35